



Lecture 9 and 10

Adsorption

By
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Introduction



Adsorption is the process by which a substance, originally present **in one** phase, is removed from that phase by *accumulation at the interface* between that phase and a separate (solid) phase.

Definitions

- **Adsorbate or solute**: the material being adsorbed (e.g., 2,4,6-trichlorophenol).
- **Adsorbent**: the solid material being used as the adsorbing phase (e.g., activated carbon).

Porous materials, like activated carbon, alumina or silica gel have “huge” surface areas (per g). This quantity is called “**specific surface area**” of solids, A_{spec} , and is a characteristic of solid materials.

In principle adsorption can occur at any solid-fluid interface such as;

- **Gas-solid interface** (as in the adsorption of a VOC on activated carbon);
- **Liquid-solid interface** (as in the adsorption of an organic pollutant on activated carbon).



Applications of adsorption

- i) stabilization,
- ii) cleaning,
- iii) heterogeneous catalysis including catalytic converters in cars,
- iv) separations like drying,
- v) surface modification and change of properties like **foaming** or **rheological** ones.
- vi) measurement of the surface area of powders and other solid surfaces or in heterogeneous catalysis and novel high surface area materials (which can sometimes be higher than $100 \text{ m}^2\text{g}^{-1}$ and, on occasion, even higher than $1000 \text{ m}^2\text{g}^{-1}$), e.g. zeolites, which can be used for **selective separations (like removing pollutants)**. Many porous materials, e.g. activated carbon, alumina or silica gel and zeolites, have very high specific surface areas.

Types of Adsorption

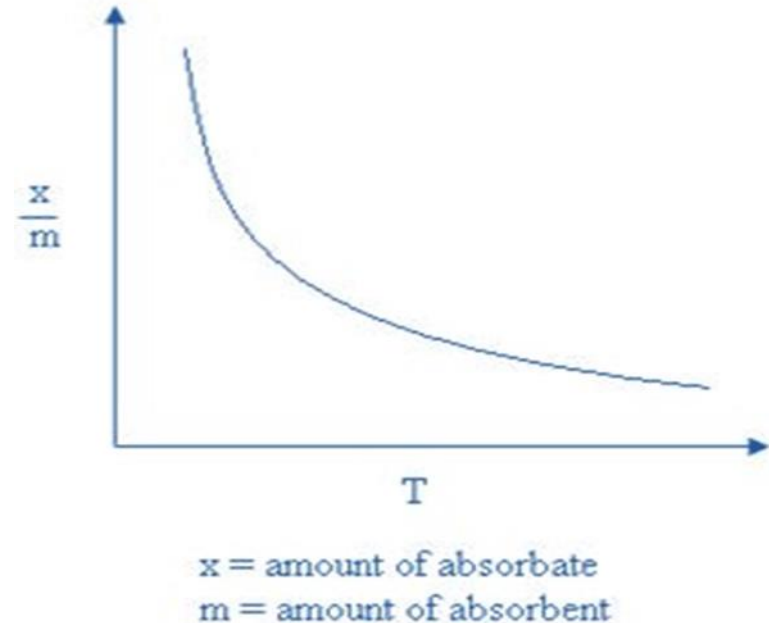


Forces of attraction exist between adsorbate and adsorbent which can be due to **Van der Waal forces** of attraction (weak) or **due to chemical bonds** (strong).

On the basis of type of forces of attraction, we have **Physical Adsorption** or **Chemical Adsorption**.

1. Physical Adsorption or Physisorption

When the weak Van der Waal forces of attraction exist between adsorbate and adsorbent, the process is Physical Adsorption or Physisorption. It takes place at low temperature below boiling point of adsorbate. **As the temperature increases, the process of Physisorption decreases.**



Physical Adsorption vs.
Temperature graph



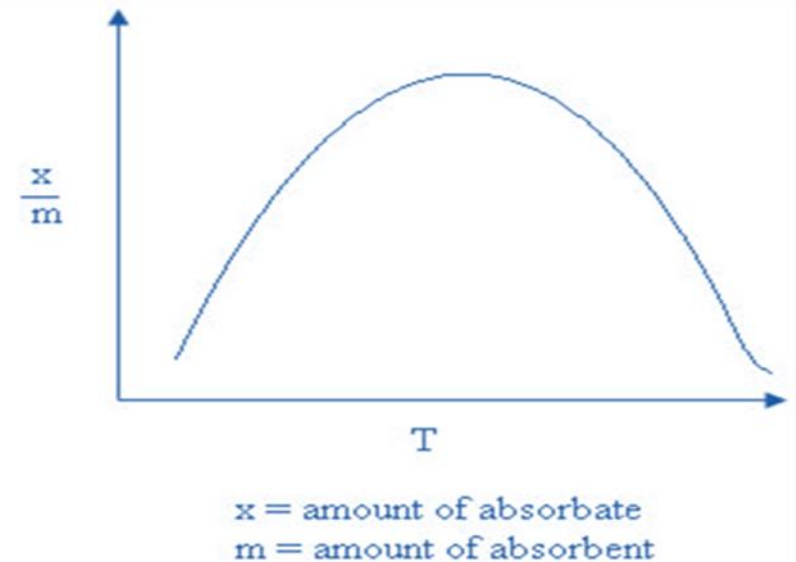
2. Chemical Adsorption or Chemisorption

When the force of attraction existing between adsorbate and adsorbent are chemical forces of attraction or chemical bond, the process is Chemical Adsorption or Chemisorption



It can take place at all temperature. With the increase in temperature, chemisorption first increases and then decreases.

Chemical Adsorption vs.
Temperature Graph



Physical adsorption

Weak van der Waals-type forces

Fast to reach equilibrium,

Reversible

Multilayer (BET) adsorption

Low heat of adsorption, $<10 \text{ kcal mol}^{-1}$

Chemical adsorption

Strong forces, like chemical bonds

Slow,

often irreversible

Monolayer (Langmuir) adsorption

5–100 kcal mol



Adsorption Isotherms



The adsorption equilibrium relates q to C . The equilibrium is a function of the temperature.

Thus, the adsorption equilibrium relationship at a given temperature is typically referred to as *adsorption isotherm*, i.e.:

$$q = f(C)$$

where: q = mass of species adsorbed/mass of adsorbent (i.e., concentration of adsorbable species in solid adsorbent); C = equilibrium concentration of adsorbable species in solution

Types of Adsorption Isotherms

Different adsorbates-adsorbents exhibit different types of equilibrium relationships (i.e., the function $q = f(C)$ may take different mathematical forms). It has been found that for most of the cases of importance in wastewater treatment the function $q = f(C)$ takes the form of one of the following isotherms:

- Langmuir isotherm
- Brauner-Emmet-Teller (BET) isotherm
- Freundlich isotherm



Adsorption using the Langmuir equation

Langmuir adsorption theory is based on the assumption that the **adsorption rate** is proportional to the concentration in the solution and the fraction of free non-covered area of the adsorbent.

The desorption rate, on the other hand, is considered to be proportional to the fraction of surface covered by adsorbate molecules. At equilibrium, the adsorption and desorption rates are equal.

The theory is based on many simplifying assumptions:

- “ideal” (clean, smooth, homogeneous, non-porous) surface;
- existence of active centres on the surface; each centre (site) can be free or occupied by an adsorbed molecule; and only one molecule per active centre;
- no interactions between adsorbed molecules, i.e. homogeneous surface;
- all sites are equivalent (same heat of adsorption);
- no solute– solvent interactions and solute and solvent have same size;
- dilute solutions; monolayer coverage.
- adsorption is a reversible process
- the adsorbed molecules do not move on the surface of the adsorbent.
- the enthalpy of adsorption is the same for all molecules independently of how many have been adsorbed



Gas – solid adsorption using the Langmuir equation



Langmuir equation is often given as the fraction Θ (uppercase theta) of occupied active centres (or volume per mass of solid) over the total active centres (or the volume of monolayer) against pressure. The equation and its linear form are shown in Equation 1.1:

$$\Theta = \frac{V}{V_m} = \frac{B.P}{1 + B.P} \Rightarrow \frac{P}{V} = \frac{P}{V_m} + \frac{1}{B.V_m} \quad 1.1:$$

The volume V is expressed in, for example, cm^3g^{-1} (solid),

P is the pressure,

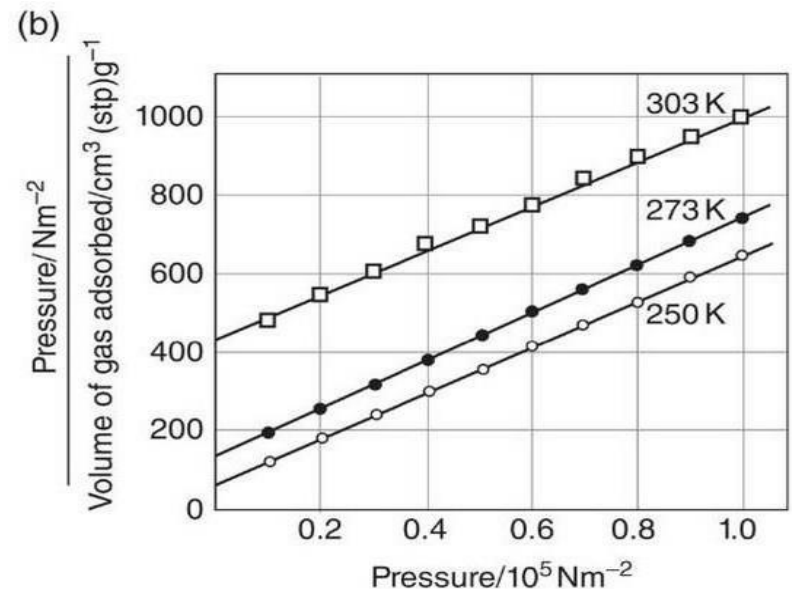
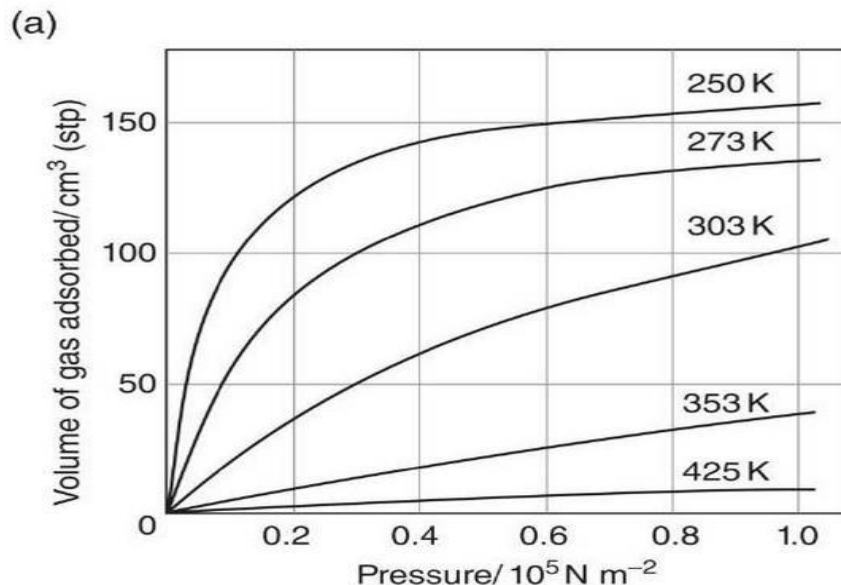
while V_m and B are adjustable parameters, respectively, the volume of a monolayer and the affinity parameter. The higher the B -parameter is, the higher is the affinity of a gas for the solid.



Note that a plot of P/V versus P gives a straight line.
The slope is $1/V_m$ and the intercept is $1/BV_m$.



At low pressures, Langmuir approaches the Henry equation ($V = BV_m$). Both parameters of the Langmuir's equation, B (affinity) and V_m (monolayer coverage), **must be estimated from experimental data**, based on the linear plot shown in Equation 1.1



Adsorption isotherms for ammonia on charcoal: (a) as plots of the volume of gas (per mass of solid), V , as a function of pressure (P) at constant temperature (T) and (b) in linearized Langmuir plots.



Adsorption of solutions using the Langmuir equation



When we have adsorption from solutions, the adsorption typically increases with increasing concentration and with decreasing temperature

The Langmuir equation for solutions is given below (in its original and linear forms):

Original Langmuir form:

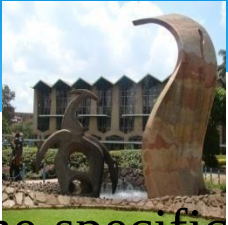
$$\Gamma = \frac{\Gamma_{\max} k C}{1 + k C}$$

In linear form:

$$\frac{C}{\Gamma} = \frac{C}{\Gamma_{\max}} + \frac{1}{k \Gamma_{\max}}$$

where $A_{\text{spec}} = N_A A_0 \cdot \Gamma_{\max}$

Typical C and Γ (uppercase for gamma) units are respectively mol L⁻¹ and mol g⁻¹, while A_{spec} is given in m²g⁻¹. If $\Gamma_{\max}$ are known then the area occupied by a molecule (A_0) can be calculated.



Notes on Langmuir parameters



The specific surface area of the solid, A_{spec} , can be estimated when the area of the adsorbed molecule is known (A_0). Alternatively, when the specific surface area of the solid is known (from other techniques, e.g. BET adsorption of gases like nitrogen on the same solid), we can estimate the area occupied by each adsorbed molecule e.g. of surfactant.

The inverse of k ($=1/k$) is the concentration at which one half of the surface is occupied by the solute.

Γ_{max} is the adsorption at full coverage (obtained above the CMC (critical micelle concentration) for surfactant solutions).

Langmuir parameters

The other parameter of Langmuir, k , is a measure of the affinity of the solute for the surface, i.e. how much the solute likes the surface. The higher the parameter k is, the higher the slope and thus the higher the affinity.



Example



The following data refer to the adsorption of dodecanol (from solution in toluene) onto a sample of carbon black. The specific adsorption area of the carbon black is $105 \text{ m}^2 \text{ g}^{-1}$ and is measured using the technique of the adsorption of nitrogen on the solid.

Equilibrium concentration (mol L^{-1})	Amount adsorbed ($\mu\text{mol g}^{-1}$)
0.012	24.1
0.035	50.4
0.062	69.8
0.105	81.6
0.148	90.7

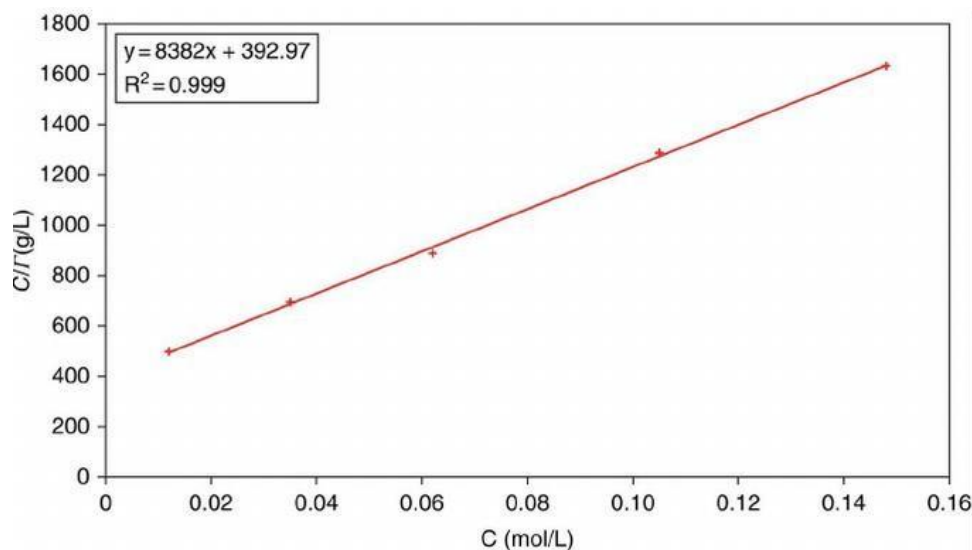
Show that the data fit the Langmuir adsorption isotherm equation and calculate (in $\text{\AA}^2$) the area occupied by each adsorbed dodecanol molecule at limiting adsorption.

Solution

We will use the Langmuir equation in the form of [Equation](#)

We plot the concentration/adsorption versus the concentration (the latter in mol L^{-1}). We use the data provided in the table and express the ratio of concentration/adsorption in units of g L^{-1} (values: 498, 694, 888, 1287 and 1632, respectively for the five points provided).

The plot is indeed linear indicating that the adsorption of dodecanol follows the Langmuir



The “maximum adsorption” = $1/\text{slope (of linear plot)} = 1/8300 = 0.00012 \text{ mol g}^{-1} = 120.5 \text{ } \mu\text{mol g}^{-1}$.

From the value corresponding to the monolayer, we can calculate the area occupied by each adsorbed dodecanol molecule using the equation:

$$A_{\text{spec}} = \Gamma_{\text{max}} N_A A_0$$

Thus:

$$105 = 120.5 \times 10^{-6} \times 6.0223 \times 10^{23} A_0$$

which gives the value of the dodecanol molecule area as $145 \times 10^{-20} \text{ m}^2$ or $145 \text{ } \text{\AA}^2$.

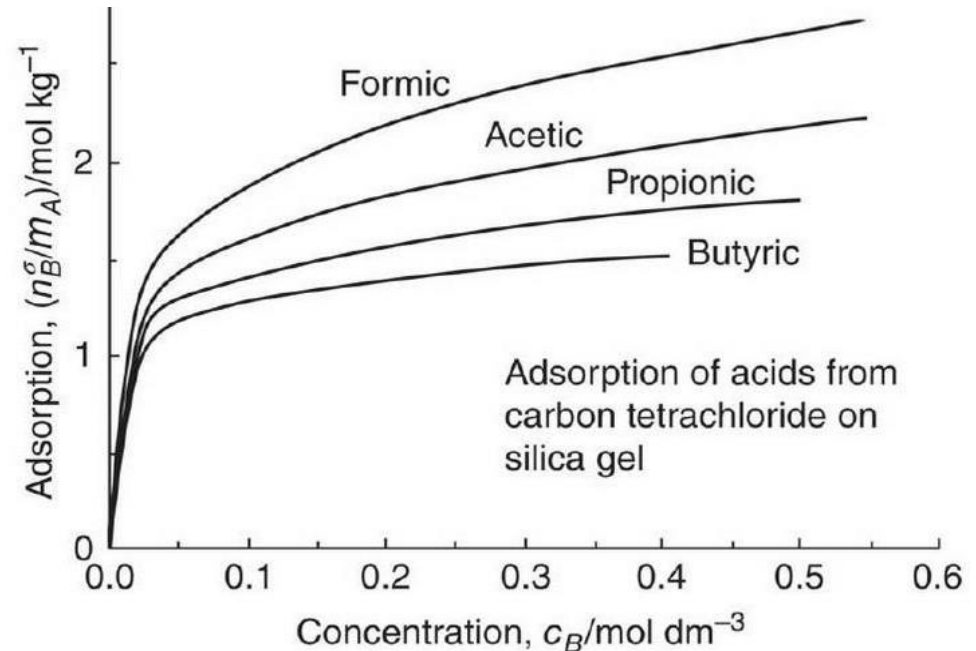


Adsorption from solution – the effect of solvent on adsorption



The adsorption of stearic acid on carbon black reaches different limiting values in different solvents; ranging from 10 mmol kg^{-1} when benzene is the solvent up to more than 40 mmol kg^{-1} when cyclohexane is the solvent

This is because the effect of solvent and the competition of solute and solvent for sites on the solid surface



The adsorption of four fatty acids on a polar surface (silica gel) from a non-polar solution (carbon tetrachloride).



Adsorption of gases on solids using the BET equation

Stephen Brunauer (1903–1986), Paul Emmett and Edward Teller (1908– 2003):- **adsorption theory**



Their purpose was to derive an equation resembling Langmuir's adsorption, which, however, can be used for multilayer physical adsorption as well, and leads to Langmuir and Freundlich (another adsorption equation) at specific limiting values. The final expression is given by Equation below;

$$\frac{P}{V(P_o - P)} = \frac{1}{CV_m} + \frac{C-1}{CV_m} \frac{P}{P_o} \quad (1.2)$$

V_m is again the volume of monolayer coverage, which is simply calculated as $1/(\text{slope} + \text{intercept})$.

The parameter C of BET is a measure of the difference in the heat of adsorption of the 2nd, 3rd and so on layer compared to the heat of the creation of the first layer.

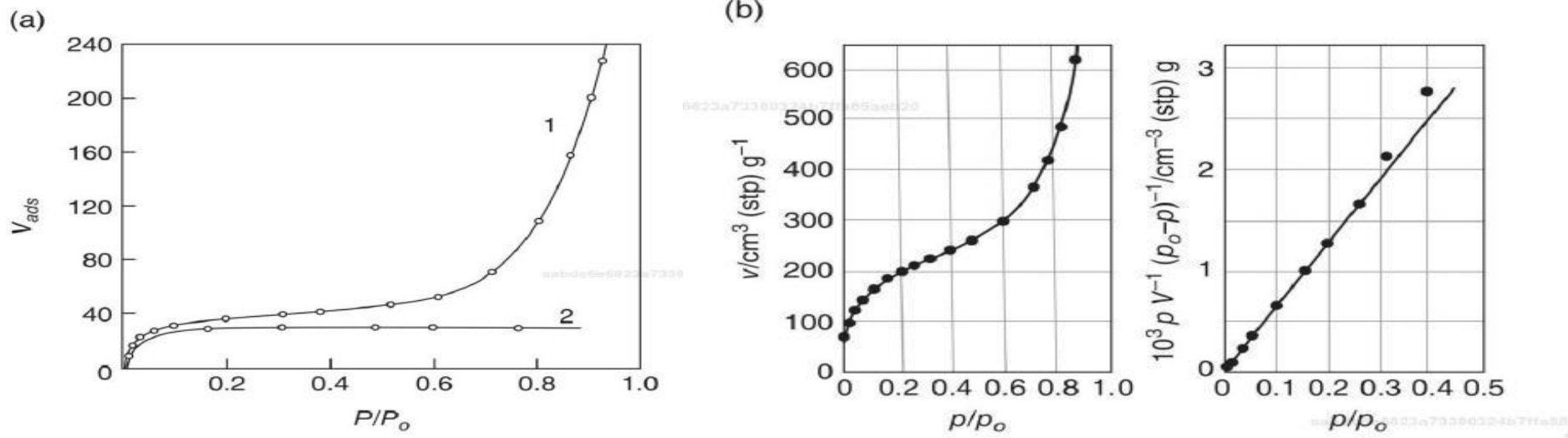
With the exception of the first layer, all other layers are assumed to have the same heat of adsorption. The first layer is characterized by a much higher heat of adsorption compared to the other layers.



Adsorptions not described by Langmuir's theory especially the **physical multilayer adsorption of gases on solids** can be best described by the BET (Brunauer– Emmett– Teller) theory.



BET can also be written in a linear form, for reduced pressures (P/P_o) up to about 0.4, and yield values of the monolayer coverage and of the specific solid surface area



(a) Comparison of two adsorptions: that of **nitrogen on silica** is a **physical BET adsorption** at 77 K (curve 1), while **oxygen's adsorption on charcoal** at 150 K (curve 2) is a **chemical Langmuir adsorption**. Notice that the y-axis is the pressure divided by the saturation pressure. (b) Adsorption isotherm **for nitrogen on silica gel** in the **linearized form of the BET equation**



The BET equation has found widespread use in adsorption studies. For example, **the model has been used for the characterization of porous materials, i.e. getting information about the surface area and the nature of the surface.**



Using Equation 1.3, the specific surface area of solids can be estimated from the V_m parameter of BET:

$$A_{spec} = \frac{V_m N_A}{V_g} A_o \quad (1.3)$$

Here V_g is the gas volume at standard temperature and pressure (s.t.p.), 1 atm 2 molecule (0.162 nm³ and 273 K (= 22414 cm³ mol⁻¹ or 22.414 L mol⁻¹. A_o is the area occupied by one gas molecule (0.162 for the often used N₂ at 77 K, 0.138 nm² for Ar and 0.195 nm for Kr). Nitrogen is the standard gas used in BET measurements of the specific surface area of solids.



Moreover, when the specific surface area of a solid is known (using the technique presented here) then we can estimate, from solution adsorption data, the area occupied by a surfactant or a polymer on the same solid, A_o .

The BET theory has also its limitations:

- *Absence of lateral interactions between adsorbed molecules;*
- *Valid up to relative pressures (P/P_o) of 0.4;*
- *Not always good agreement with calorimetric measurements;*
- *Extension to n -layer for porous materials yield complex equations that cannot be expressed in linear form.*



Example: Adsorption based on BET equation is used.



The following data have been obtained for the adsorption of nitrogen on TiO_2 at 77 K:

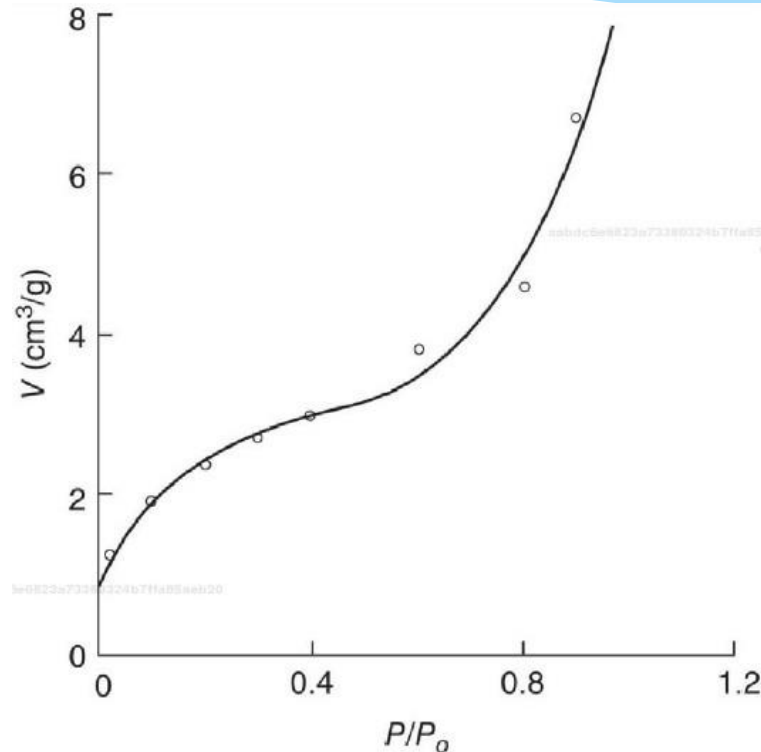
P/P_o	0.02	0.05	0.1	0.2	0.3	0.4	0.6	0.8	0.9
volume of gas ($\text{cm}^3 \text{ g}^{-1}$)	1.2	1.5	1.9	2.4	2.7	3.0	3.8	4.6	6.7

1. Show that the adsorption isotherm is likely to be represented by the BET equation.
2. Estimate using the BET equation:
 - a. the volume for monolayer coverage V_m (in $\text{cm}^3 \text{ g}^{-1}$);
 - b. the specific surface of solid TiO_2 in $\text{m}^2 \text{ g}^{-1}$ if the area occupied by one nitrogen molecule is $16.2 \times 10^{-20} \text{ m}^2$.



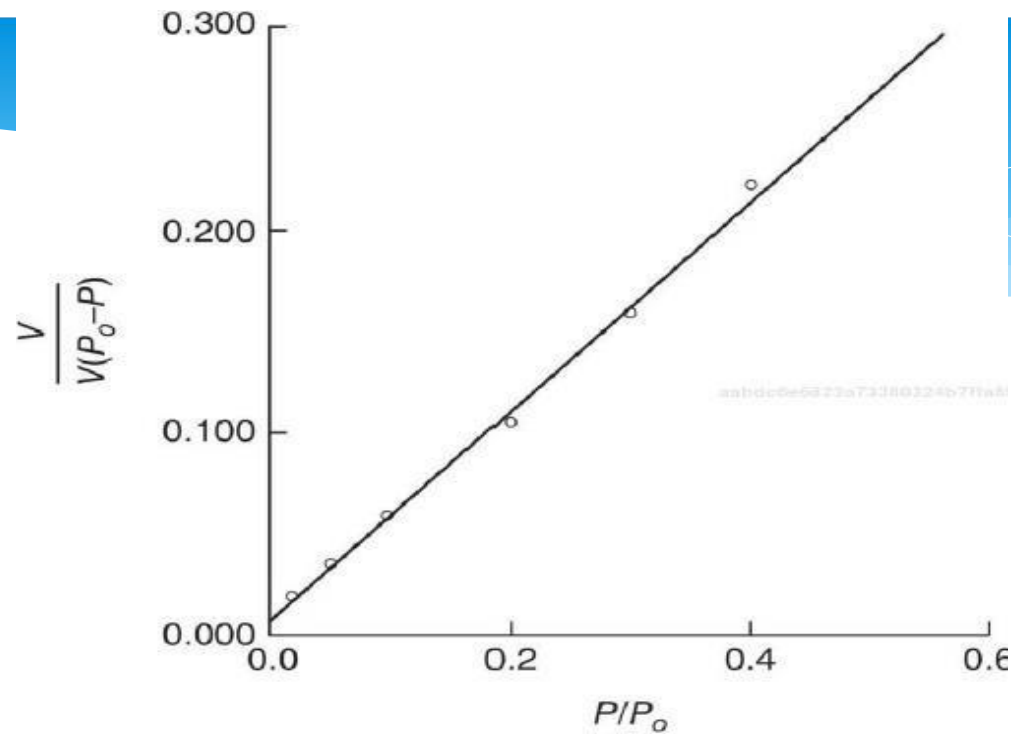
Solution

The V vs P/P_o plot clearly shows that the adsorption of N_2 on TiO_2 has the BET shape



According to the BET theory, the plot of $P/V(P_o - P)$ versus P/P_o should be linear. If we consider all the points provided in the above table, i.e. up to very high pressures, the resulting plot is *not* linear. However, the BET equation should be constructed based only on the points up to reduced pressures of about 0.4, i.e. up to sixth or seventh point. Indeed the plot of $P/V(P_o - P)$ versus P/P_o is linear up to reduced pressure 0.4 (approximately even up to the seventh point of reduced pressure 0.6).

The straight line is shown in the plot below.



The straight line is represented by the equation:

$$P/[V(P_o - P)] = 0.0052999 + 0.526907(P/P_o)$$

(Notice that as only the ratio P/P_o is given, all sides of the fraction $P/[V(P_o - P)]$ should be divided by P_o .)

Then, as known, the volume for monolayer coverage $V_m = 1/(\text{slope} + \text{intercept}) = 1/(0.526907 + 0.0052999) = 1.8789 \text{ cm}^3 \text{ g}^{-1}$.

The specific surface area of solid TiO_2 is then calculated based on this value of V_m as:

$$A_{sp} = \frac{V_m N_A}{V} A_o = \frac{1.8789 \times (6.0225 \times 10^{23}) \times 16.2 \cdot 10^{-20}}{22414} = 8.178 \text{ m}^2 \text{ g}^{-1}$$